

1 Information Technology

1. **Introduction to AI** : Definition, brief history, and scope, Problem Solving and Search with AI, Evolution of AI Technologies, Introduction to Neural Network, Machine Learning, Deep Learning, Generative AI; AI tools and platforms, Legal, ethical, and social implications of AI.
2. **Overview of Information Technology** : Information Concepts and Processing, Block Schematic of digital computer and its working, Introduction to computer hardware and software, Overview of Operating System, Basics of computer networks and internet, Information Security basics, IT Act and digital ethics.
3. † **Intro to DBMS** : Definition, purpose of databases, Real-life examples and applications, Data vs. Information, File System vs. Database System, Limitations of traditional file systems, Advantages of DBMS (data integrity, security, redundancy reduction), Relational Databases, Definition and functions, Relational Model Basics, Tables, rows (tuples), columns (attributes), Keys: Primary, Foreign key, Basic SQL syntax, DDL: CREATE, DROP, ALTER, DML: SELECT, INSERT, UPDATE, DELETE
4. **C Language**

2 Maths

1. **Calculus** :
 - (a) Partial Derivatives
 - (b) Jacobian
 - (c) Maxima and Minima of functions in two variables
 - (d) Expansion of functions by Taylor's and Maclaurin's series of two variables
 - (e) Differential equations
 - (f) Beta and gamma function
 - (g) Multiple integrals
2. † **Matrices** : Rank of a matrix by Echelon form and Normal form, Inverse of matrices by Gauss- Jordan method, System of linear equations: Solving system of Homogeneous and Non- Homogeneous linear equations, Eigen values, Eigen vectors with their properties and applications, Cayley Hamilton theorem: Finding inverse and power of a matrix by Cayley Hamilton Theorem.
3. **Fuzzy sets** : Introduction to Fuzzy Sets, Classical sets vs Fuzzy sets, Crisp set characteristics: Membership Functions, Definition and representation, Types of membership functions, Properties of membership functions: Normality Convexity Support, Operations on Fuzzy Sets: Union, Intersection, and Complement, Applications of Fuzzy Sets.
4. **Evolution of Indian Mathematics and Its Contributors** : Ancient Era: Baudhayana: Ancient Indian version of the Pythagoras theorem, Aryabhata Varāhamihira, Medieval Era: Brahmagupta Bhaskara II, Madhava of Sangamagrama, Modern era: Ramanujan, Shanti Swaroop Bhatnagar, Harish Chandra.

3 Physics

1. **Lasers** : Photon absorption & emission, role of S N Bose, components of lasers, optical resonator, Einstein's A & B coefficients, Population inversion, Ruby and He-Ne lasers, contribution of CKN Patel and M L Bhaumik,
2. **Fibre Optics** : NS Kapany's fiber optics, Classification of fibers, acceptance angle, numerical aperture, V-number, losses, FO sensors, FO communication Systems.
3. **Quantum Theory** : Theory of superposition, Planck's radiation formula, S Chandrashekar and idea of black hole the Bose in Boson, Compton's effect, pair production, de Broglie's and Heisenberg's hypotheses, Schrodinger's wave equation, interpretation of wave function, particle in a 1-D potential well.
4. **Special Theory Of Relativity** : Postulate of STR, time dilation, length contraction, twin paradox, Relativistic velocity of particles, Relativistic variation of mass, mass-energy relation
5. **Quantum Computation** : Classical Gates and operations, AND, OR, NAND, XOR Gates and operations. Unitary operations, Pauli Matrices, Quantum Logics: Hadamard, Pauli-X, Y, Z and R, CNOT, Swap Gates and their unitary operations, influence of ECG Sudharshan and GS Agarwal : Classical Gates and operations, AND, OR, NAND, XOR Gates and operations. Unitary operations, Pauli Matrices, Quantum Logics: Hadamard, Pauli-X, Y, Z and R, CNOT, Swap Gates and their unitary operations, influence of ECG Sudharshan and GS Agarwal.

4 Mechanical

1. **Introduction of IC Engines and Automobile** : Classification of IC engines, terminology used in I C engine; Otto & Diesel Cycle; their air standard efficiencies. Principal parts of IC engine, their functions and working. Introduction of electrical vehicle, components of automobile and their function like brakes, clutch, gear etc.
2. **Fluid Mechanics** : Langragian & Eulerian Method, Types of Flow and Types of motion, local and connective acceleration, continuity equation.
3. **Mechanisms And Machines** : Types of motion, links, kinematic pair, types of joints, degree of freedom, classification of kinematic pairs, kinematic chain, linkage, mechanism and structure, inversions of basic mechanisms (four- bar chain, single slider and double slider crank mechanisms) and their applications. Hook's joint, Types of gears and gear trains.
4. **Engineering Materials and Properties** : Introduction, properties, normal tensile shear and thermal stress, stress-strain diagrams for ductile and brittle materials, elastic constants and relationship between elastic constant.
5. **Engineering Drawings** : Principles of Engineering Graphics and their significance, usage of Drawing instruments, lines, lettering and dimensioning, Introduction of Projection of Line, Plane, Solid and Section of Solid, overview of orthographic and isometric projection.

5 Chemistry

1. **Water** : Introduction, Source, Types of impurities and their effects, Hardness, its expression and determination, BIS specifications for Industrial and domestic water, overview of Boiler troubles and softening methods, Numericals.
2. **Introduction to lubricants , fuels & Combustion**
 - (a) Lubricants: Introduction, Types of lubricants and principles of lubrication's, properties (test) of lubricants, numericals.
 - (b) Fuel and combustion, numericals